

Find the Percent of a Number

Lesson 4-4

Name: _____

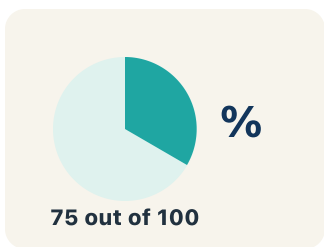
Date: _____

Class: _____

Key Vocabulary

Level 1 support

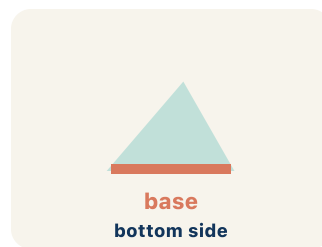
Picture first, then the word, then a plain-language meaning. Say each word out loud.



30% means 30 out of 100

Percent

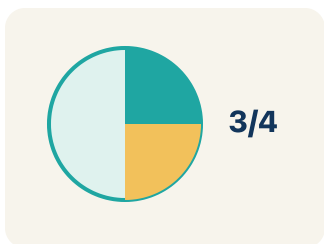
A way to compare a number to 100, shown with the % sign.



In '25% of 80,' the base is 80

Base

The whole amount you take a percent of.



25% of 80 = 20; the part is 20

Part

The piece you get when you take a percent of a number.

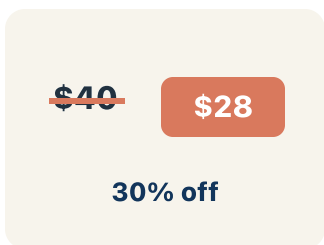
$$x + 2 = 7$$

balanced with =

part = percent $\times$ base

Equation

A math sentence with an equal sign showing both sides are the same.



\$40 shirt at 25% off: discount = \$10

Discount

Money taken off the first price to make it cheaper.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the percent of a number using an equation or a model.

1 Notice

It's Discount Day at the arcade!

2 Key idea

I can find the percent of a number using an equation or a model.

3 Words I'll use

Percent

Base

Part

Equation

Discount



Think About It

- What information do you need to find the discount amount?
- How is finding 25% of 60 different from finding 40% of 80?
- Is there more than one way to find a percent of a number?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What is 20% of 150?

A. 30

B. 20

C. 35

D. 75

Step 1

$20\% = 0.20$.

Step 2

Multiply: $0.20 \times 150 = 30$.



Answer: A. 30

 We do — together

Solve this one with your class using the same steps.

Problem: What is 50% of 84?

- A. 42
- B. 50
- C. 34
- D. 168

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: What is 10% of 320?

- A. 32
- B. 3.2
- C. 320
- D. 64

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

On Discount Day the claw machine is 25% off its 60-ticket price. In '25% of 60,' which number is the base (the whole), and what part are you trying to find?

Level 1 support

Start here: The base is 60 tickets and the part is the 25% discount we need to find.

 **Need a hint?**

Sentence starters (optional)

The base is ____ and the part I want is ____.

La base es ____ y la parte que quiero es ____.

I need to find ____% of ____.

Necesito encontrar el ____% de ____.

Word bank: percent base part discount

Listen for: Students identify 60 as the base, the 25% discount as the part to find, and connect 'of' to the base.

Level 2

Without computing exactly, will the discount be more or less than half of 60? Use a benchmark to estimate and explain.

- The discount will be ____ than half because ____.
- I estimate about ____ tickets because ____.

EXPLORE

What steps did you follow every time to find the percent of a number in the discount table?

Level 1 support

Start here: I changed the percent to a decimal, then multiplied it by the base to get the part.

 **Need a hint?**

Sentence starters (optional)

I changed ____% into the decimal ____.

Cambié ____% al decimal ____.

Then I multiplied ____ x ____ = ____ (the part).

Luego multipliqué ____ x ____ = ____ (la parte).

Word bank: percent base part equation

Listen for: Students describe the equation $\text{part} = \text{percent} \times \text{base}$, converting the percent to a decimal first (e.g., $0.40 \times 80 = 32$).

Level 2

Compare finding 25% of 60 with the decimal 0.25 versus dividing 60 into 4 equal parts.

Why do both methods give the same part?

- Both give ____ because 25% is the same as ____.
- I prefer the ____ method because ____.

CONNECT

The arcade donates 12% of Saturday's 1,500 ticket sales. The manager says 180 tickets, a student says 18. Who is right, and how do you know?

Level 1 support

Start here: 12% of 1,500 is 0.12 times 1,500, which is 180 tickets.

 Need a hint?

Sentence starters (optional)

The ____ is correct because 12% of 1,500 = ____ x 1,500 = ____.

El/La ____ tiene razón porque 12% de 1,500 = ____ x 1,500 = ____.

The mistake was using ____ instead of ____.

El error fue usar ____ en lugar de ____.

Word bank: percent base part equation

Listen for: Students compute $0.12 \times 1,500 = 180$ and explain the student likely used 0.012 (divided by 1,000) instead of 0.12.

Level 2

Estimate first: 10% of 1,500 is easy to find. Use that to check whether 180 is reasonable for 12%, and explain your reasoning.

- 10% of 1,500 is ____, so 12% should be a little ____.
- 180 is reasonable because ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words percent, base, part, and equation.

1. Kernel Sentence subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.

Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.

Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Percent matters in math

Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Percent matters in math because ____.

Porcentaje importa en matemáticas porque ____.

but
pero

Percent matters in math, but ____.

Porcentaje importa en matemáticas, pero ____.

so
entonces

Percent matters in math, so ____.

Porcentaje importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about percent.
Di un hecho verdadero sobre percent.

Percent ____.

Question *Pregunta*

Ask a question about percent.
Haz una pregunta sobre percent.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about percent.
Muestra entusiasmo sobre percent.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with percent.
Dile a un compañero qué hacer con percent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The base is ____ and the part I want is ____.

La base es ____ y la parte que quiero es ____.

I need to find ____% of ____.

Necesito encontrar el ____% de ____.

I changed ____% into the decimal ____.

Cambié ____% al decimal ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A jacket costs \$60 and is 25% off. How much is the discount?

- A. \$15
- B. \$25
- C. \$45
- D. \$20

Show your work:

2. What is 150% of 40?

- A. 60
- B. 45
- C. 20
- D. 140

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A class has 32 students. 25% are in the math club, 50% are in the reading club, and 12.5% are in both. How many students are in each group? Can the total of all three groups be more than 32? Explain why or why not.

Sentence starter: There are ____ in math club, ____ in reading club, and ____ in both. The total ____ be more than 32 because ____.

Show your work:

Reflect — Exit Ticket

What is 45% of 360?

- A. 162
- B. 145
- C. 180
- D. 126

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $42 - 50\% = 0.50$. Multiply: $0.50 \times 84 = 42$. (Or simply divide by 2: $84 \div 2 = 42$.)
2. **You Try (notes):** A. $32 - 10\% = 0.10$. Multiply: $0.10 \times 320 = 32$. (Or move the decimal one place left.)
3. **Try It 1:** A. $\$15 - 0.25 \times 60 = \15 off.
4. **Try It 2:** A. $60 - 1.5 \times 40 = 60$.
5. **Exit Ticket:** A. $162 - 45\% = 0.45$. Multiply: $0.45 \times 360 = 162$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).